TECHNICAL DESIGN DOCUMENT

Smart EV Battery Management System (Smart BMS)

System architecture, high-efficiency subsystems, and safety control design

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1. Executive Summary

The Smart EV Battery Management System (BMS) is a high-efficiency hardware-software framework designed for real-time state estimation, safety enforcement, and active cell balancing in electric vehicle battery packs.

The proposed architecture integrates active charge transfer, solid-state electrical isolation, low-latency cell monitoring, and predictive battery diagnostics to improve usable energy, extend service life, and reduce safety risks. Thermal propagation prevention requires additional mechanical, thermal, and pack-level measures; electrical isolation alone cannot guarantee it.

2. System Architecture & Component Design

2.1 Top-Level Architectural Block Diagram

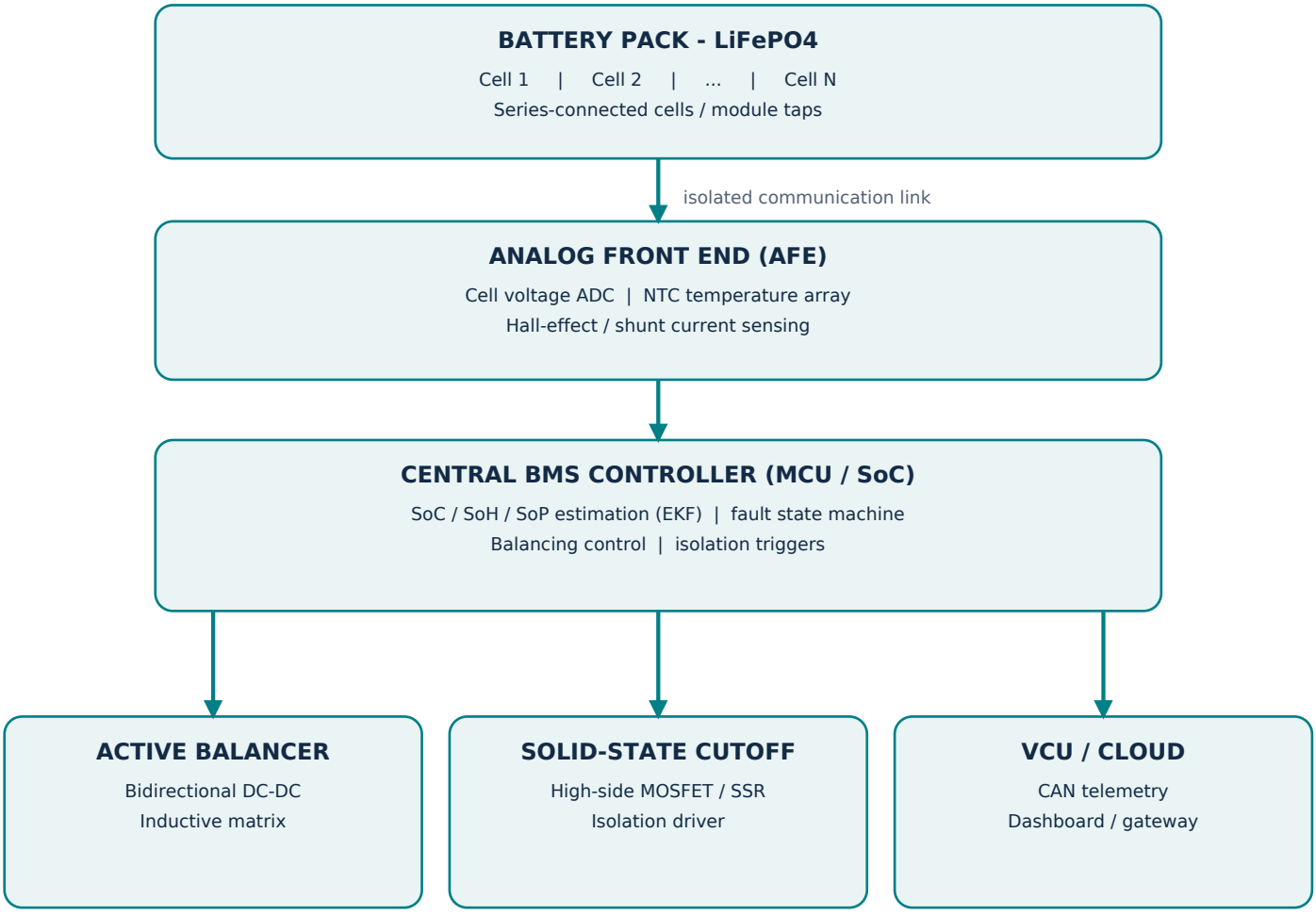

Figure 1. Functional architecture and principal sensing, control, and actuation paths.

The AFE digitizes cell and pack measurements; the central controller estimates battery state, evaluates faults, commands balancing and isolation, and communicates operating limits and diagnostics to the vehicle controller.

3. High-Efficiency Subsystem Specifications

3.1 Active Charge Transfer (Balancing Subsystem)

Topology: Bidirectional flyback or switched-capacitor/inductive matrix. Energy is transferred from higher-charged cells to lower-charged cells or a shared pack bus instead of being dissipated in resistors. Design target: greater than 88% charge-transfer efficiency, subject to topology, operating point, and validation.

3.2 Real-Time Cell Monitoring Layer

Voltage: 12- to 16-bit ADC with target sampling intervals of 10 ms or less and target accuracy of ±1 mV (requires complete AFE error-budget verification). Current: Hall-effect sensing or calibrated shunt with suitable isolation and range. Temperature: distributed NTC thermistors at representative cell and busbar locations.

3.3 Solid-State Thermal Runaway Isolation

Monitor absolute temperature, temperature rate-of-rise (dT/dt), cell overvoltage/undervoltage, and current. A high-side MOSFET or suitably rated solid-state switching assembly provides an electrical isolation path. Design target: less than 1 ms electronic cutoff command-to-switch response under specified conditions. Fault interruption capability, inductive energy handling, safe operating area, and contactor coordination must be validated; disconnecting a module does not itself stop an ongoing thermal runaway.

3.4 Key Design Targets

Parameter	Proposed target / approach
Cell chemistry	LiFePO4
Balancing efficiency	> 88% (validation target)
Cell-voltage acquisition	12-16 bit ADC; ≤ 10 ms target interval
Voltage measurement accuracy	± 1 mV target; system error budget required
Electronic cutoff response	< 1 ms target; dependent on load/fault case
Vehicle communications	CAN 2.0B / gateway telemetry

4. Software Architecture & State Machine

The controller begins in self-test, enters normal operation only after required checks pass, applies derating during warning conditions, and latches critical faults into a safe isolation state according to the approved recovery policy.

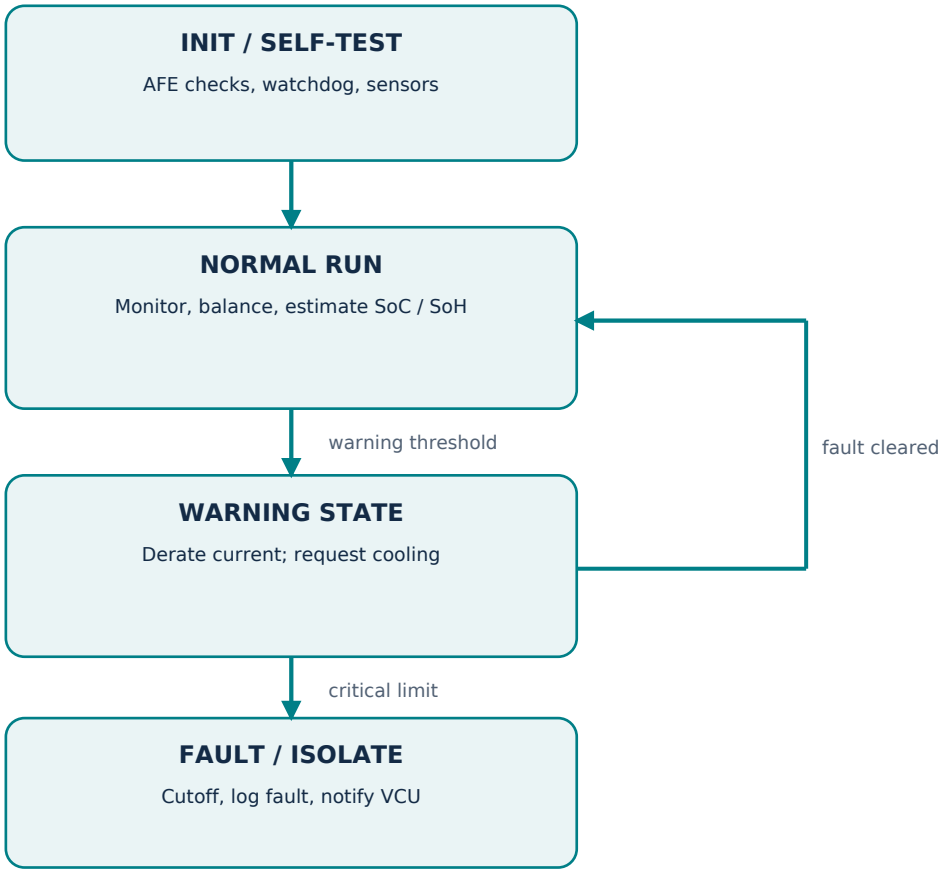

Figure 2. Simplified operating-state transitions. Recovery from a critical isolation fault requires a separate, validated reset strategy.

4.1 Control and Fault Responsibilities

INIT / SELF-TEST: Verify AFE communications, sensor plausibility, memory integrity, watchdog and switching feedback. NORMAL RUN: Estimate SoC, SoH and SoP; monitor thresholds; manage balancing. WARNING: Limit allowable current and request thermal management. FAULT / ISOLATE: Inhibit switching, request/execute isolation, retain diagnostic records, and signal the VCU.

5. Verification, Safety, and Implementation Notes

Verify accuracy over temperature, EMC behavior, sensor faults, isolation diagnostics, switching transients, worst-case fault current, balancing efficiency and thermal behavior. Define fault priorities, hysteresis, debounce intervals, communications timeout behavior, and safe-state transitions in detailed requirements. Applicable vehicle and battery safety standards must be identified for the target market and pack configuration.

Engineering caveat: The numeric values in this document are proposed targets, not demonstrated product specifications. This conceptual design is not sufficient by itself for production safety approval.